Healthcare AI Hackathon

AI Triage Predictor

Smarter Emergency Care, Starting at Your Fingertips

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Team 911

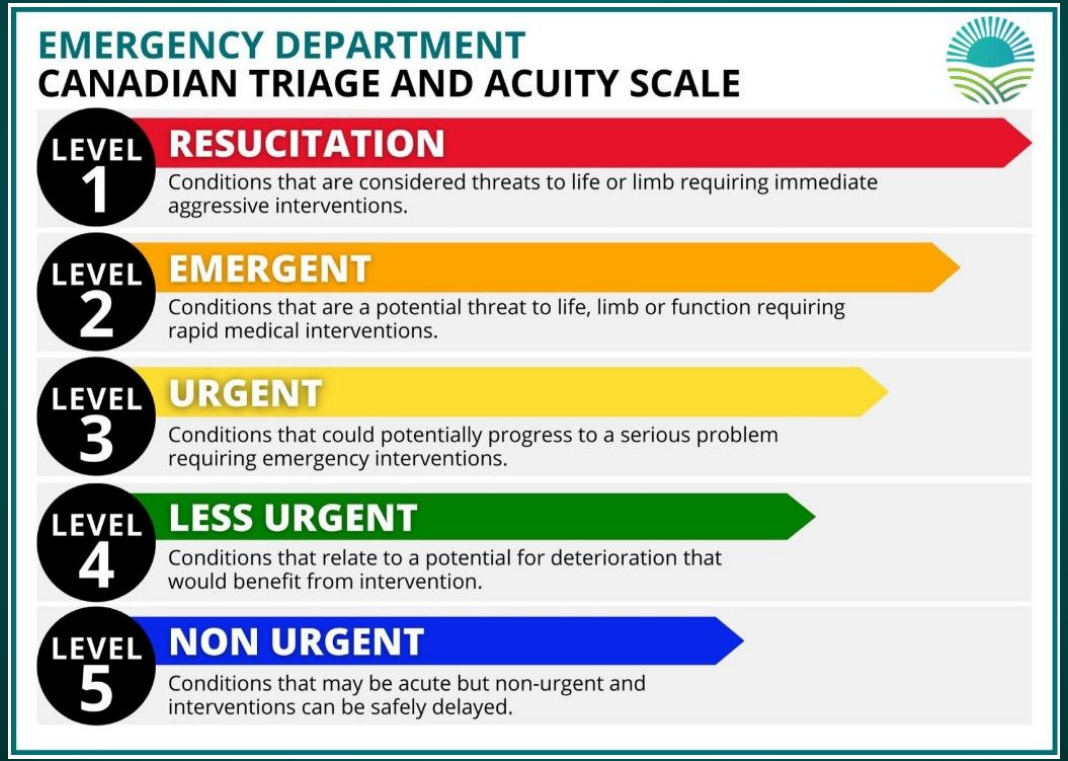
The Real Problem

Canada's ER System Is Under Pressure

- 1 family doctor for every ~1,000+ Canadians
- Average ER wait times can exceed 4–6 hours
- Patients arrive without knowing how serious their condition is
- Overcrowding leads to missed urgencies and unnecessary visits

What is CTAS?

The Scale That Decides Who Gets Help First



OUR SOLUTION

RECAP

A Virtual Triage Assistant Powered by AI

1

2

3

4

Analytics

Use the backend triage model to estimate likely CTAS level from a patient summary.

Patient summary

52 year old with shortness of breath, troponin 0.42, sodium 133, taking metoprolol

Predict triage level

Likely triage level

CTAS 4 - Less Ur...

↑ Baseline CTAS 3 - Urgent

Identified chief complaint

Shortness Of Breath (99% confidence)

Triage level	Probability (%)
CTAS 4 - Less Urgent	42.62
CTAS 5 - Non-Urgent	41.62
CTAS 3 - Urgent	11.75
CTAS 1 - Resuscitation	2
CTAS 2 - Emergent	2

Recommended where to go

General Practitioner, Walk-in Clinic, or over-the-counter care for minor issues

What this level means

Mild conditions that require medical evaluation but are not time-sensitive.

Why this destination fits

These conditions can usually be managed safely outside the hospital with simple treatments or routine follow-up.

Common examples

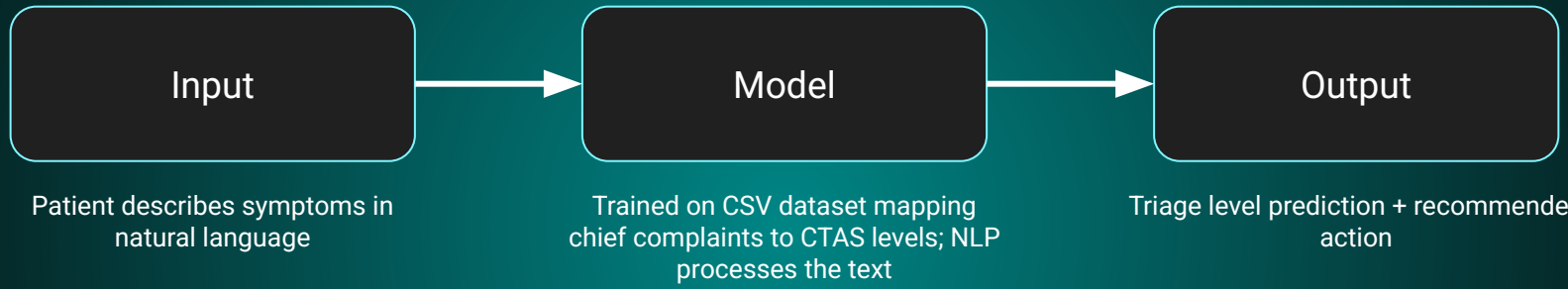
- Mild infections like ear, throat, or urinary infections
- Minor sprains
- Rashes
- Medication refills
- Mild cold or flu symptoms

Grounded in Real Clinical Data

- Large CSV dataset of real triage records
- Features include: chief complaint (plain English), vital signs indicators, final CTAS rating
- Model learns the language patterns clinicians use to assign urgency

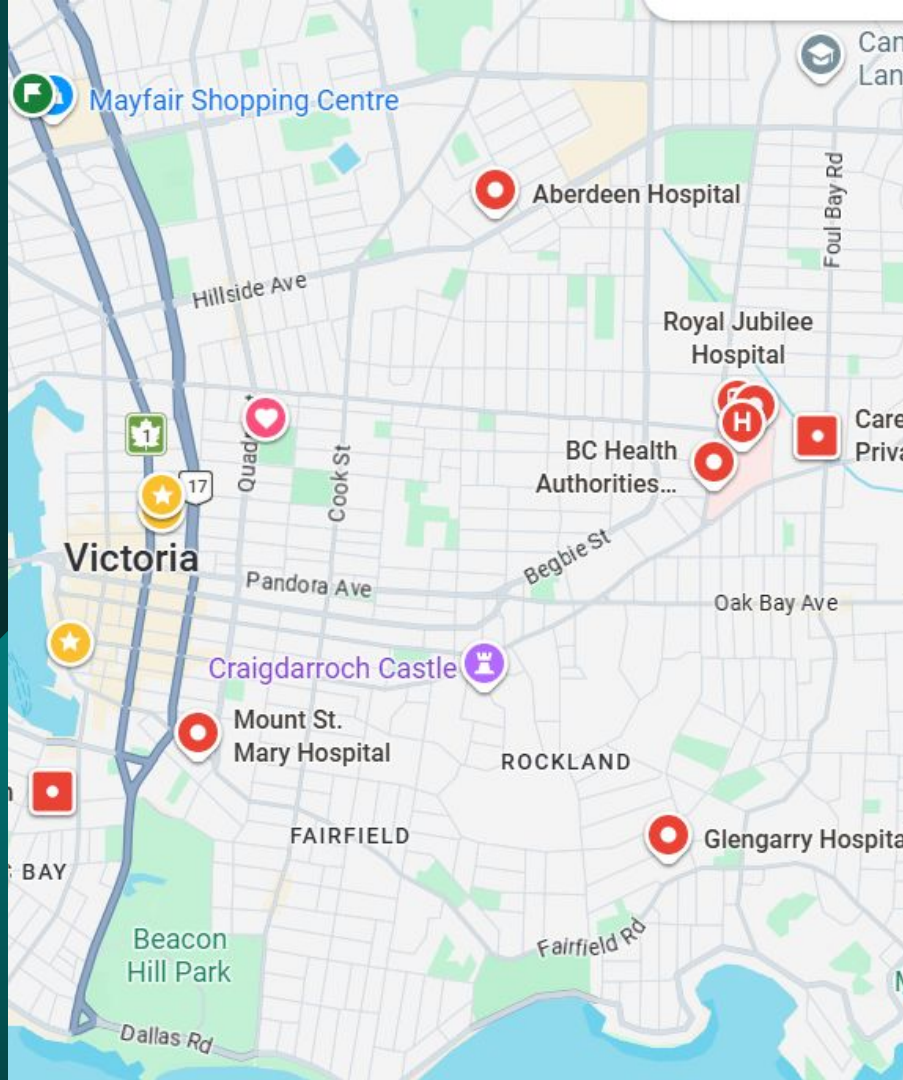
Under the Hood

How it works?



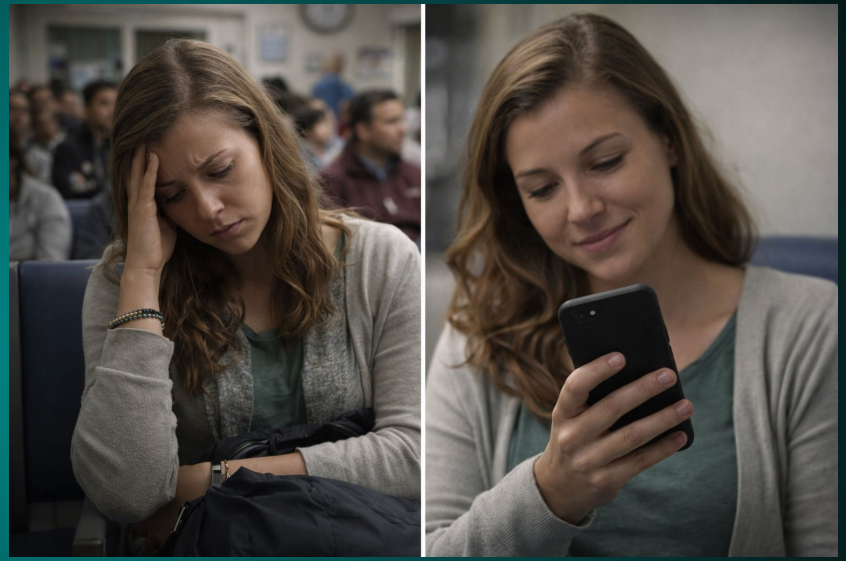
Future Considerations: Where This Goes Next

Appointment booking tied to triage level — urgent cases get earlier slots
Hospital routing — recommends the nearest ER with the shortest wait time
Crowding awareness — live data integration to redirect patients from overwhelmed ERs
Multilingual support — critical for diverse Canadian communities



WHY IT MATTERS?

- Reduces unnecessary ER visits for low-urgency cases
- Helps high-urgency patients self-identify and act faster
- Lowers patient anxiety by giving clear, actionable guidance
- Potential to reduce strain on Canada's overburdened healthcare system



Demo

Thank you!

ANY QUESTIONS?